Sruti Bapatla

She/Her | 732-915-1248 | sbapatla@andrew.cmu.edu | linkedin.com/in/sruti-bapatla | U.S. Citizen

EDUCATION

Carnegie Mellon University | GPA: 3.85/4.0 | University Honors | Pittsburgh, PA — May 2026
Bachelor of Science in Mechanical Engineering
Additional Majors in Double Bass Performance and Biomedical Engineering, Medical Device Track
Veritas University | San Jose, Costa Rica — September - December 2024

RELEVANT EXPERIENCE

Republic of Palau Ministry of HR | Technology Consultant | Koror, Palau — May 2026 - July 2026
- Creating HR dashboards through Power BI to automate report generation
- Programming an AI-driven chatbot for government website, to answer FAQs

Carnegie Mellon University | Fifth Year Scholar | Pittsburgh, PA — September 2025 - May 2026
- Launched nonprofit-sponsored capstone program connecting local organizations with CMU mechanical engineering students

Neocis | Surgical Instruments Intern | Miami, FL — June - August 2025
- Redesigned oral fiducials to accommodate an oxygen mask, considering DFMA and aesthetics
- Created and tested fixtures, utilizing Solidworks and 3d printing, to precisely place beads in pockets

Smith + Nephew | Robotics Intern | Pittsburgh, PA — June - August 2024
- Designed and tested pins utilizing bone simulations in Solidworks to determine design with the least compressive stresses
- Conducted compression and deflection testing to optimize placement of bone pins for shoulder procedures

Smith + Nephew | Manufacturing Engineering Intern | Memphis, TN — June - August 2023
- Programmed 135 knee and hip zirconium oxide implants using VEI software to validate part size and family
- Created VRA documentation and work instructions to introduce new VEI vision software to clean room

Caterpillar | Summer Engineering Intern | Peoria, IL — June - August 2022
- Created scripts using NI DIAdem to analyze and visualize engine data to better advertise engines to customers
- Automated and ran daily tests on dual-fuel engine to check engine health and diagnose turbine efficiency issues

PROJECT EXPERIENCE

Auto - Measuring Spice Dispenser — September - December 2025
- Designed and prototyped a compact refillable spice dispenser using mechanical components to dispense preset increments

Spinal Torque Device Biomedical Engineering Capstone (Sponsored by Medtronic) — September - May 2023
- Designed and manufactured a biocompatible adapter for measuring applied torque on spinal surgical instruments

Drug Eluting Stent Simulation — January - May 2023
- Simulated a drug eluting stent using COMSOL to analyze stresses on stent and drug release kinetics

Scrabble Python Game — January - May 2022
- Programmed a multi-player, multi-level online scrabble game against computer AI using Python

LEADERSHIP EXPERIENCE

Teaching Assistant, Carnegie Mellon Department of Mechanical Engineering — September 2023 - May 2026
Courses: 2D & 3D Statics & Design, Mechanical Design, Mechanical Systems Experimentation, DIY Design, Graduate Seminar
Student Leadership & Advocacy
Mechanical Engineering Student Advisory Council Leader, SWE Advocacy Chair, Jazz Ensemble Director — September 2021 - May 2026
Makerspace Student Worker — January 2024 - May 2026
Carnegie Mellon University Ambassador, Mechanical Engineering Department Ambassador — January 2022 - May 2026
Community Volunteer - Literacy Pittsburgh (English Tutor), Soup Kitchen, Animal Shelter, Community Garden — August 2023 - Present

SKILLS AND COURSES

Software: Python, MatLab, Java, Solidworks, AutoCAD, COMSOL, ANSYS, Onshape, Power BI, Adobe Illustrator
Machinery: Laser Cutter, 3D Printer, Lathe Machine, Milling Machine, Water Jet
Languages: Spanish, Telugu
Courses: Systems Thinking and Sustainability, Applied Finite Element Analysis, Communication Design, Women and Health in America

HONORS

Andrew Carnegie Scholar | 1 of 40 students selected as a senior representative for academic excellence, leadership, and service
Carnegie Mellon Senior Leadership Recognition | Selected for making significant contributions to CMU community
Mortar Board Honors Society
Pi Tau Sigma Honors Society (President)